Week 2

Machine Learning and Big Data - DATA622

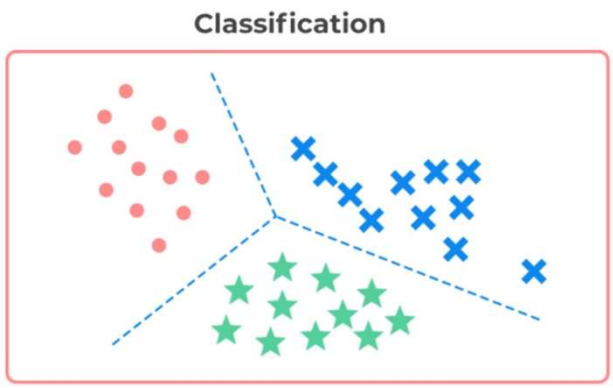
CUNY School of Professional Studies

Week 2

1. Discussion Board Week 2: Supervised vs unsupervised learning
2. Reading: Practical Machine Learning in R (PMLiR)
   - PMLiR Chapter 4: Linear Regression (60 mins)
     Chapter covers simple and multiple linear regression
   - PMLiR Chapter 5: Logistic Regression (60 mins)
     Chapter covers how logistic regression is used for classification

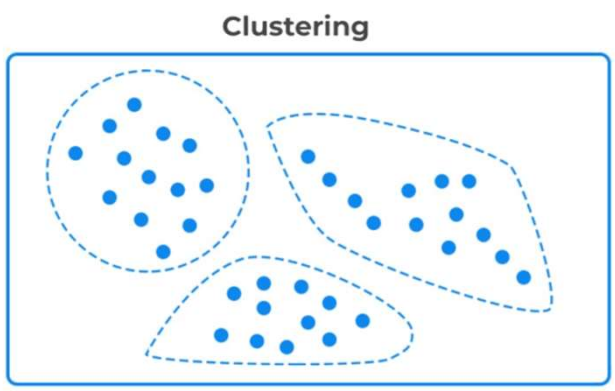
3. ML Concepts Reading & Videos
   Linear & Logistic Regression
   Read slides #1 to #28: https://web.stanford.edu/~hastie/MOOC-Slides/classification.pdf (30 mins)
   - Multinomial Regression
     Read: https://stats.oarc.ucla.edu/r/dae/multinomial-logistic-regression/ (20 mins)
   - Statistical learning and models
     Videos: View entire series please: https://www.youtube.com/watch?v=p9n2w236B48&list=PL5-da3qGB5IDvuFPNoSqheihPOQNJpzyy&index=1 (60 mins)

Supervised vs Unsupervised



Supervised learning

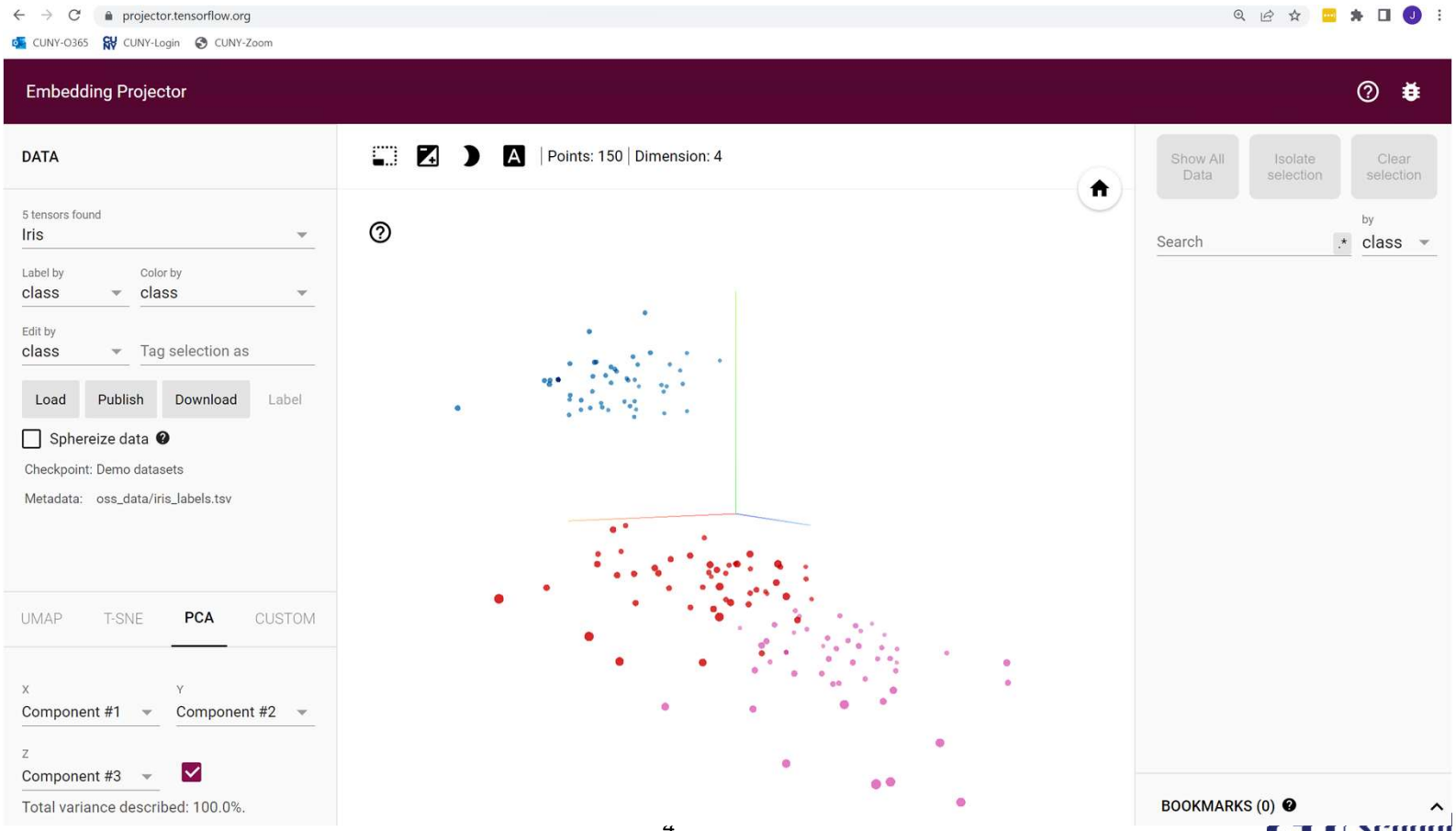
- **Supervised Learning**
  - The computer is presented with **both**
    - inputs ("features" i.e. independent variables) **and**
    - outputs ("labels") i.e. associated sample outputs, indicating the class of the observation (dependent variable)
  - The computer attempts to learn the model that maps inputs (features) to each class (labels)
  - It predicts outputs: new data inputs are classified based on the model learned



Unsupervised learning

- **Unsupervised Learning**
  - The computer is given only inputs ("features" i.e. independent variables)
  - The computer attempts to cluster data based on similarity/dissimilarity of inputs
  - It cannot predict outputs – as it was never given sample outputs

Visualizing data

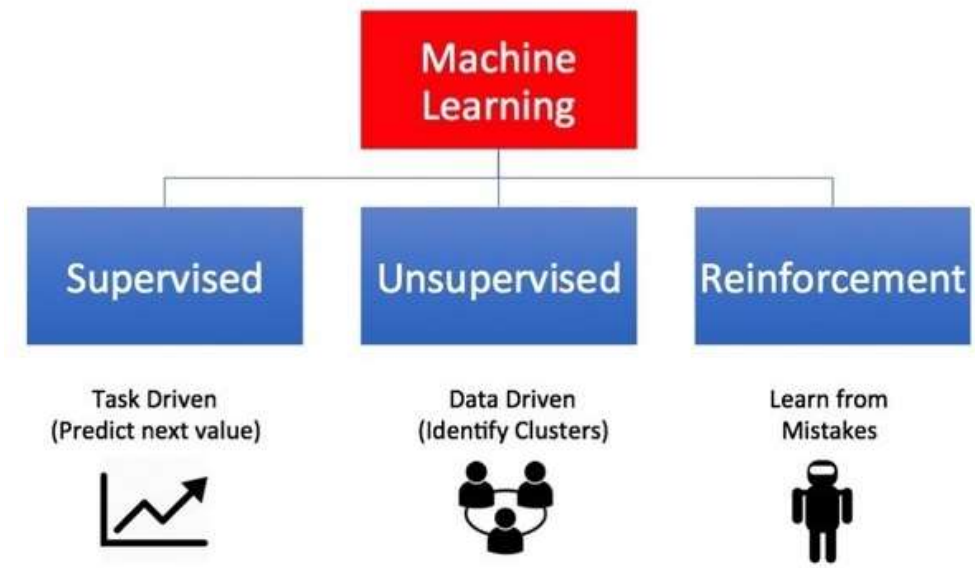


Machine Learning vs Statistical Learning

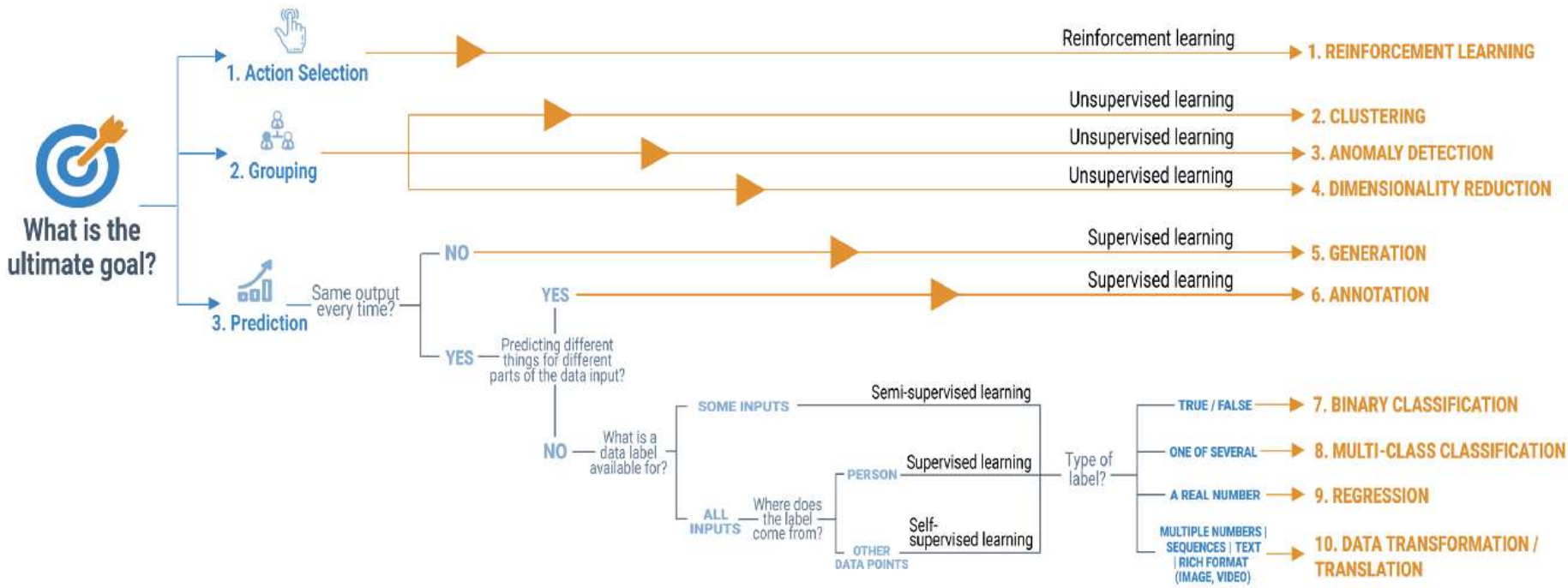
MACHINE LEARNING	STATISTICAL LEARNING
Subfield of Artificial Intelligence	Subfield of mathematics
Uses algorithms	Uses equations
Requires minimum human effort; is automated	Requires a lot of human effort
Can learn from large data sets	Deals with smaller data sets
Has strong predictive abilities	Gives a best estimate: you gain some insights into one thing, but it's of little or no help with predictions
Makes predictions	Makes inferences
Learns from data and discovers patterns	Learns from samples, populations, and hypotheses

Source: *An introduction to Statistical Learning*

Types of Machine Learning



Types of Machine Learning



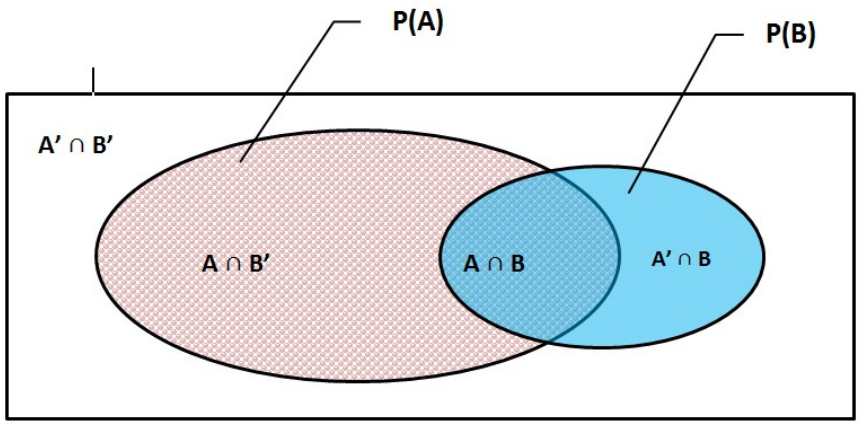
Bayes Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

P(A)= The probability of A occurring P(B)= The probability of B occurring

P(A? B)=The probability of A given B P(B? A)= The probability of B given A

P(A? B))= The probability of both A and B occurring



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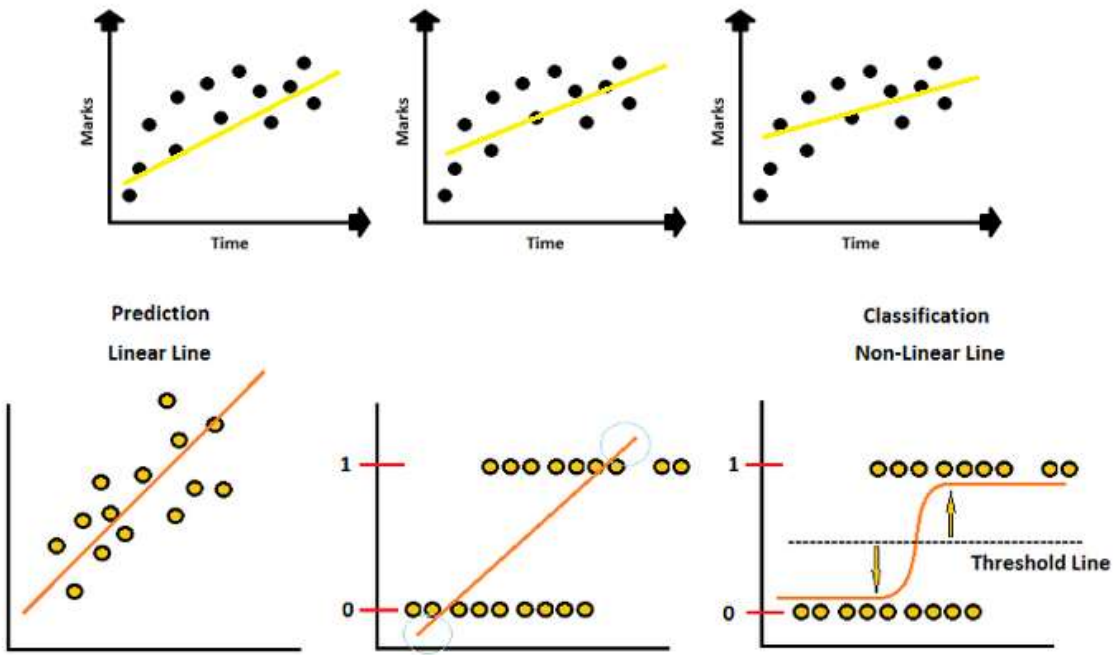
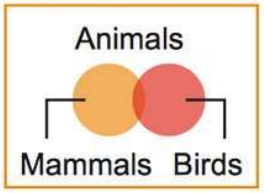
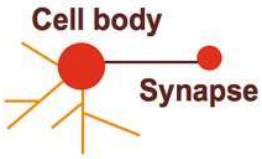
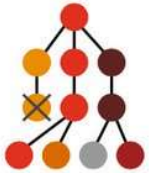
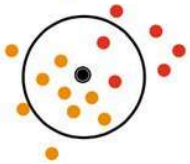
Linear vs Logistics Regression



Five Tribes of Machine Learning

What are the five tribes?

Symbolists	Bayesians	Connectionists	Evolutionaries	Analogizers
				
Use symbols, rules, and logic to represent knowledge and draw logical inference	Assess the likelihood of occurrence for probabilistic inference	Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons	Generate variations and then assess the fitness of each for a given purpose	Optimize a function in light of constraints ("going as high as you can while staying on the road")
Favored algorithm Rules and decision trees	**Favored algorithm** Naive Bayes or Markov	**Favored algorithm** Neural networks	**Favored algorithm** Genetic programs	**Favored algorithm** Support vectors

Source: Pedro Domingos, *The Master Algorithm*, 2015